Project title:

ChatGpt Client –ChatGpt-Style Interface with Flutter &Riverpod Using Mobile Application Development

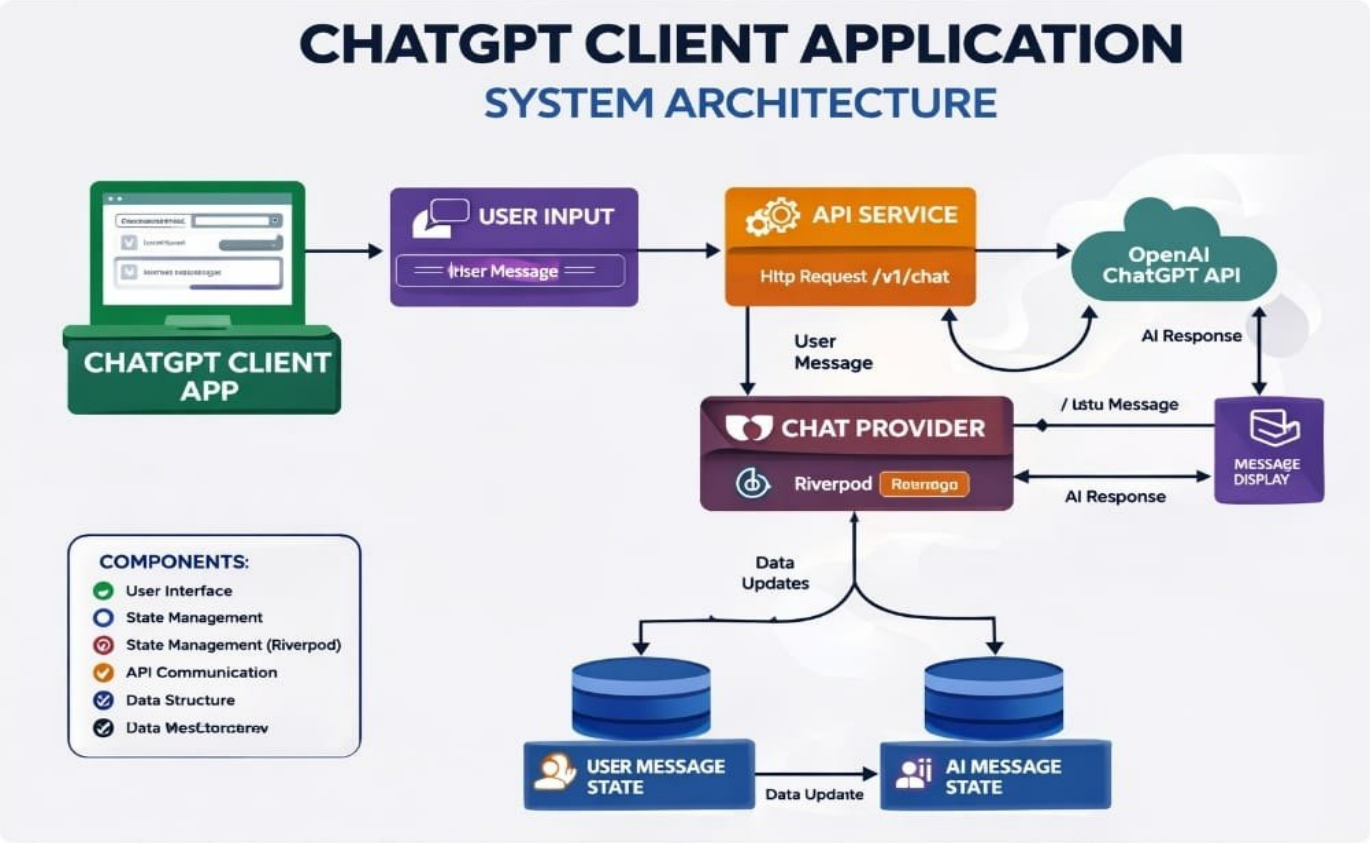
Introduction:

- This Project is a ChatGpt-Style mobile app built using Flutter and Riverpod for state management
- It is developed using Flutter, across-platform framework for building modern UI applications.
- The app users to interact with an AI model through a clean chat interface.
- The system supports real-time message exchange between user and AI.
- The UI design focuses on simplicity ,responsiveness,and user-friendly experience.
- Messages are displayed in a chat bubble format,similar to popular messaging apps.
- It demonstrates the use of modern mobile app architecture and best practices.
- IT can be extended with features like chat history,themes,voice input and authentication.
- This project is help in understanding API intergration and UI design in Flutter.

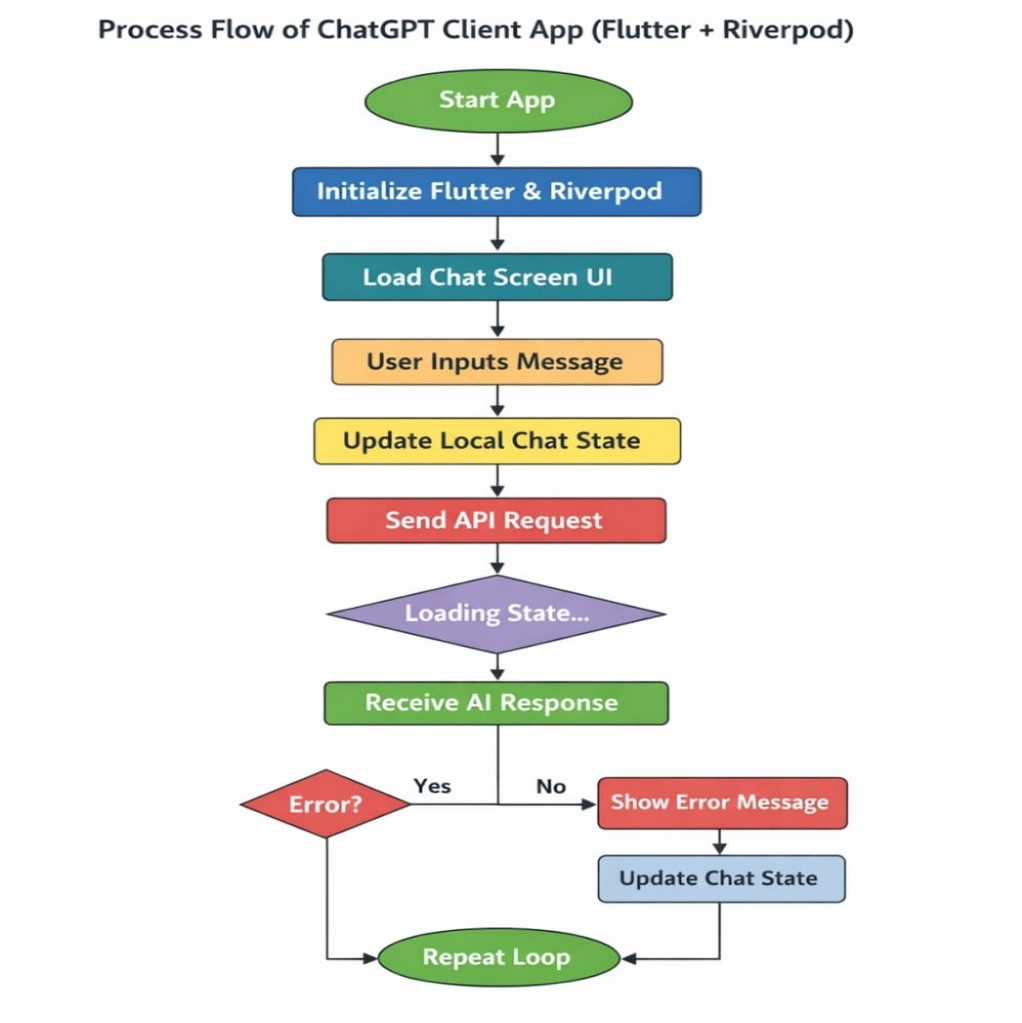
Objectives:

- To develop a chatgpt-style mobile application using
- Flutter
- To implement a responsive and interactive chat user interface.
- To use Riverpod for efficient state management in the application.
- To design a clean,user-friendly and modern UI/UX.
- To integrate AI API for generating intelligent responses.
- To manage chat data dynamically with in the app.
- To enable real-time communication between user and AI system.

System Architecture:



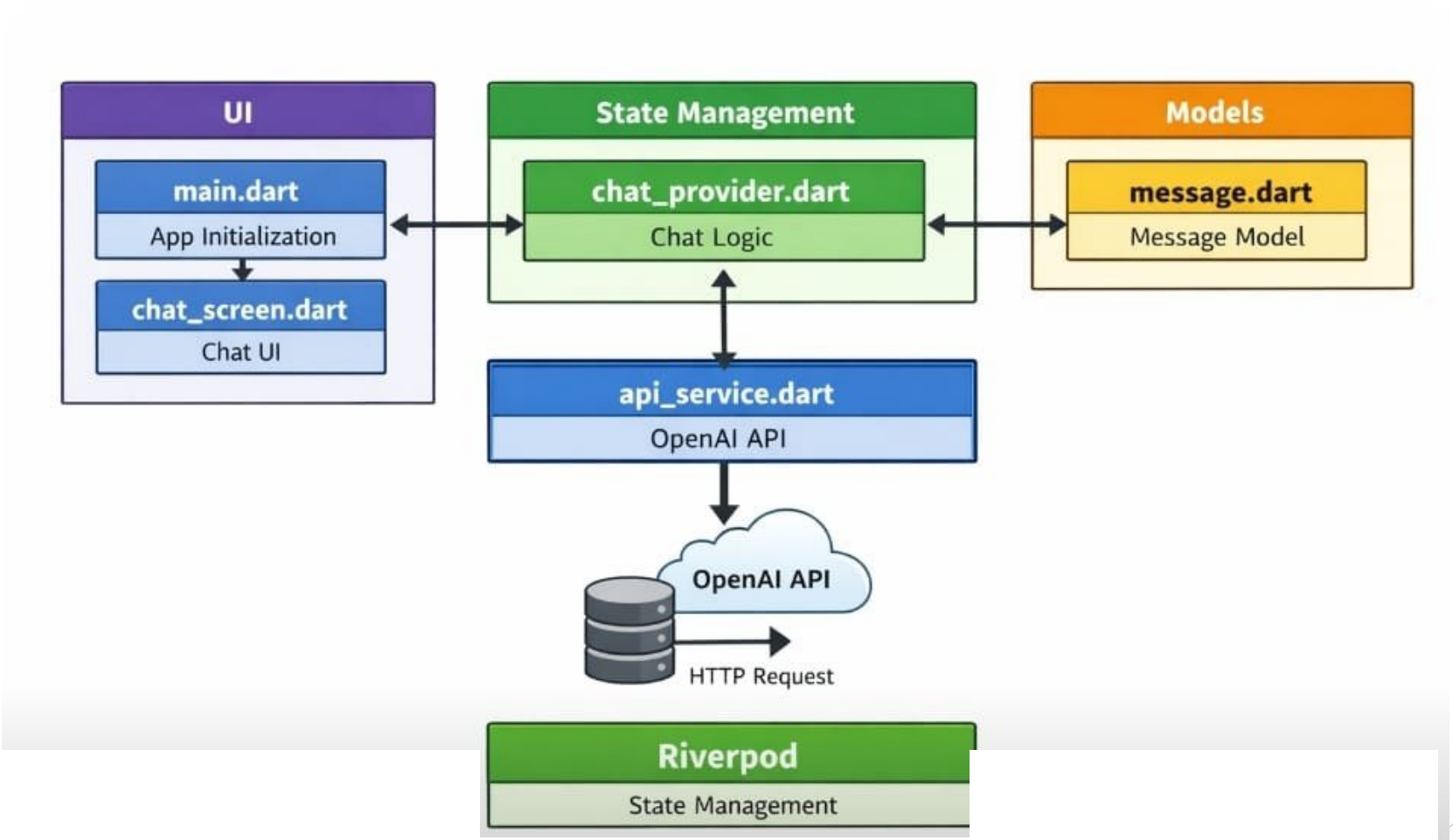
Process Flow:



Source Code Overview:

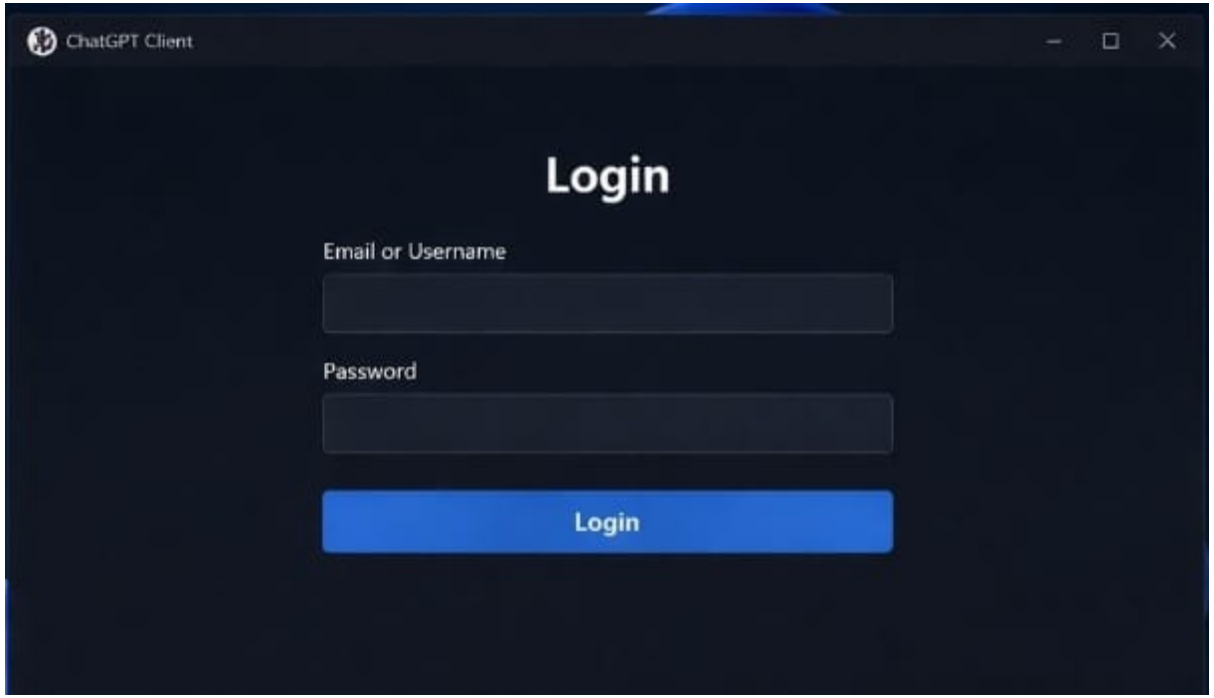
* `main.dart`
  Starts the app
  Loads the main screen
*`message.dart`
  Stores message details
  Includes text, sender, and time
* `chat_provider.dart`
  Manages chat messages
  Adds user and AI messages
  Updates the screen automatically
* `api_service.dart`
  Sends user message to API
  Receives AI response
* `chat_screen.dart`
  Shows chat messages
  Contains input box and send button
* `Riverpod`
  Manages app state
  Keeps UI updated
* `HTTP Package`
  Used for API communication
  `Flutter Widgets`
  Used to build UI

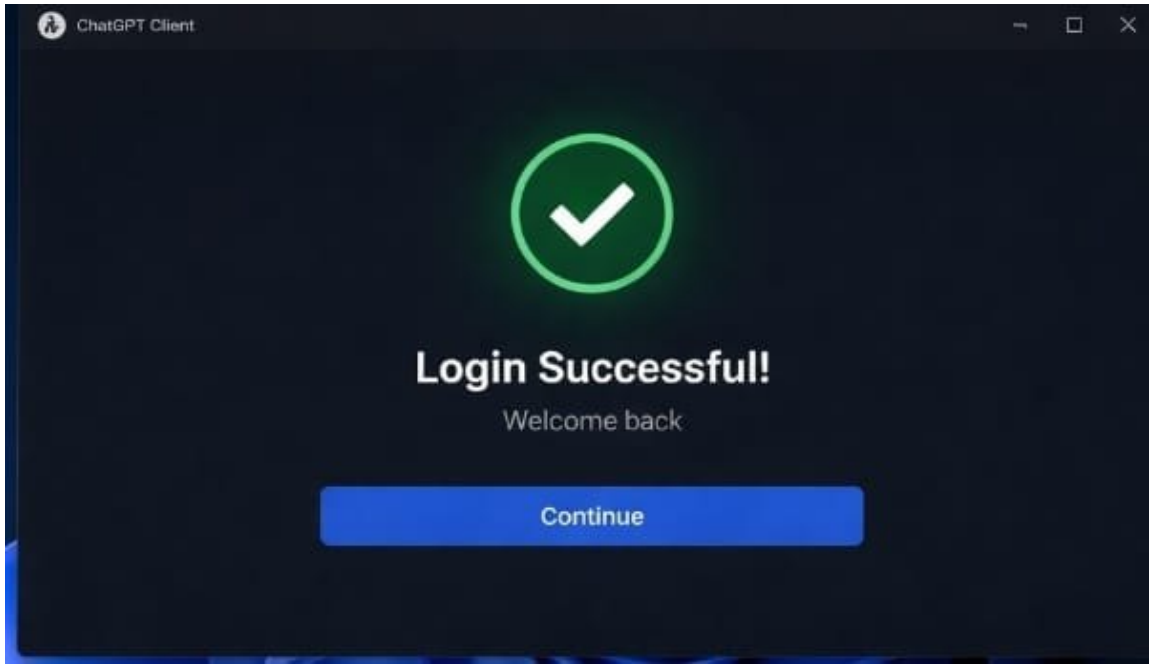
Source Code Components :

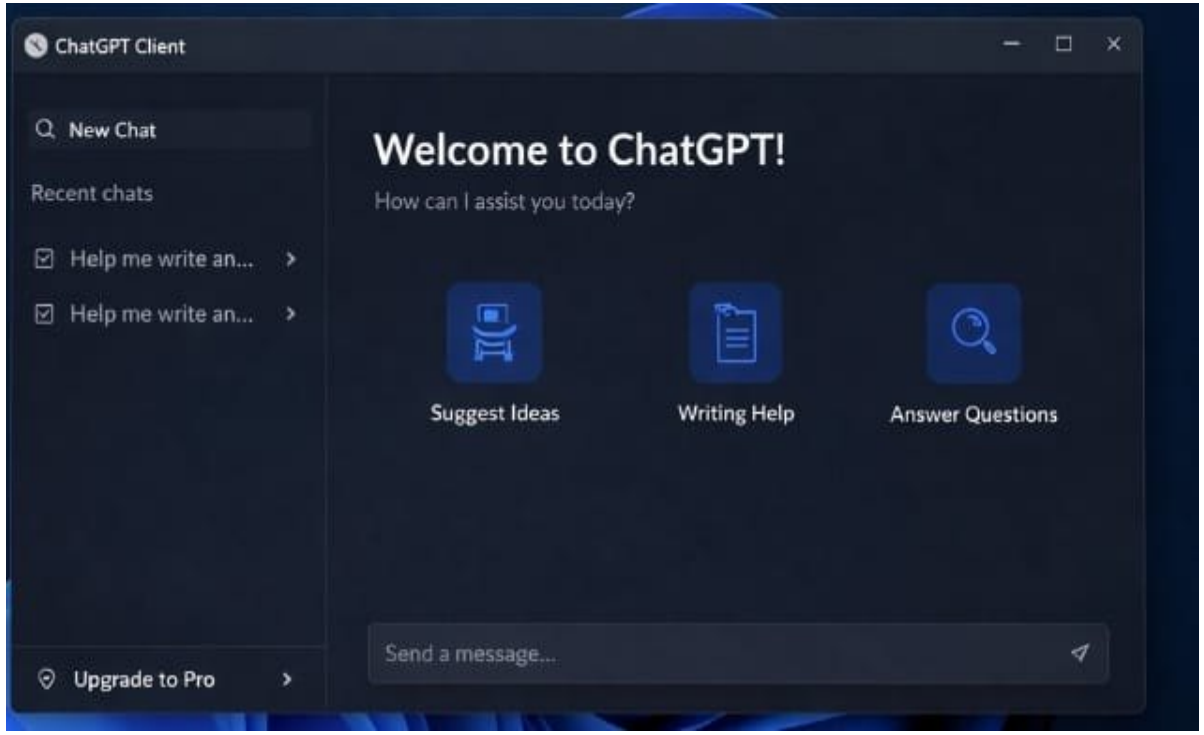


Login Module:

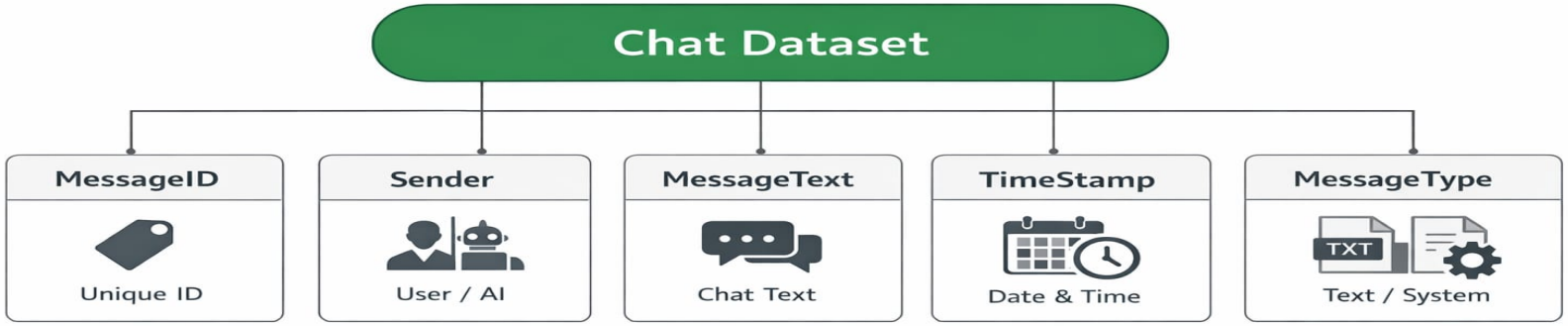
- User enters email and password
- System checks login details
- If details are correct, login is successful
- User is taken to chat screen
- If details are wrong, error message is shown



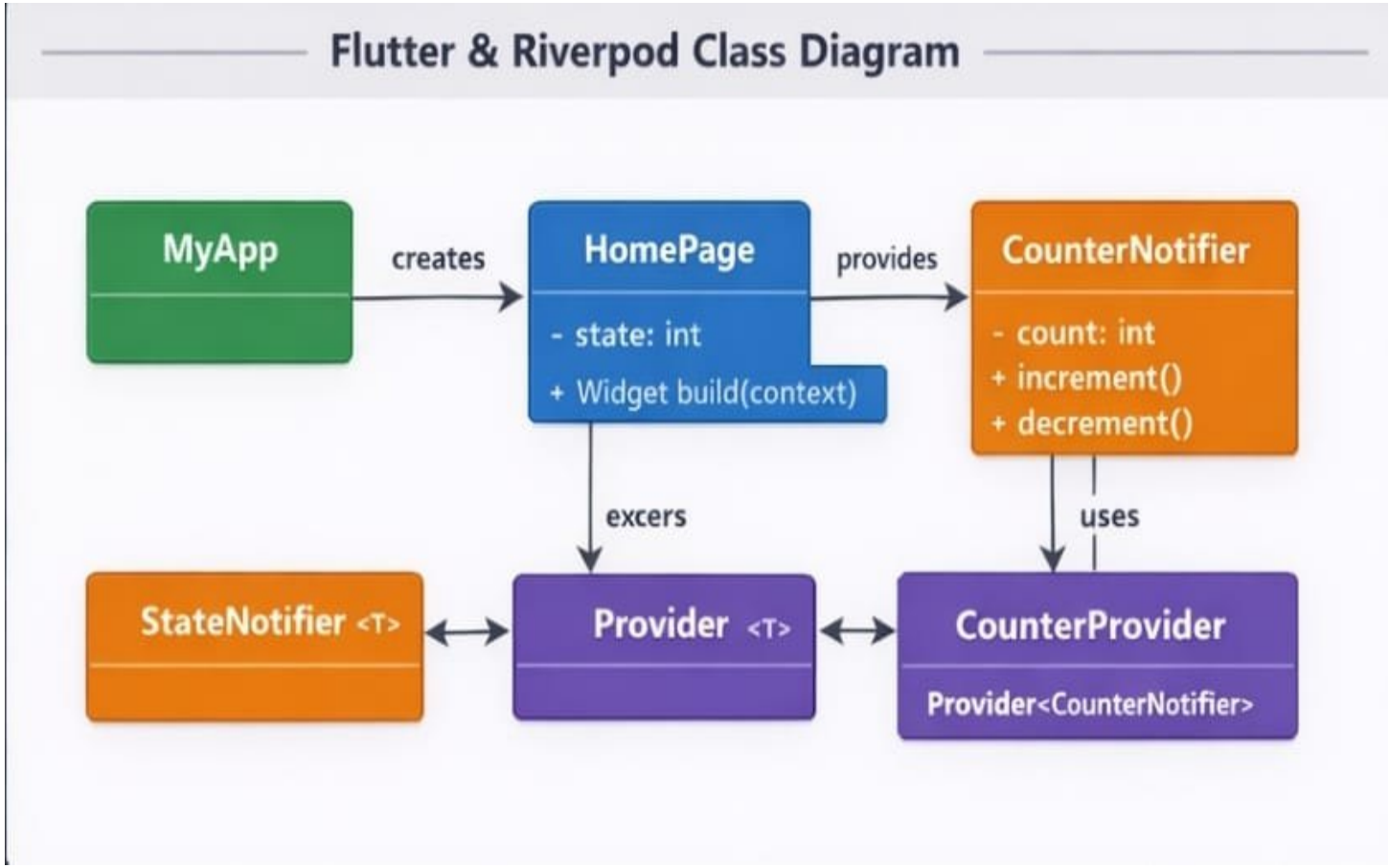

ChatGPT Client
Login Successful!
Welcome back
Continue


ChatGPT Client
New Chat
Recent chats
Help me write an...
Help me write an...
Upgrade to Pro
Welcome to ChatGPT!
How can I assist you today?
Suggest Ideas
Writing Help
Answer Questions
Send a message...

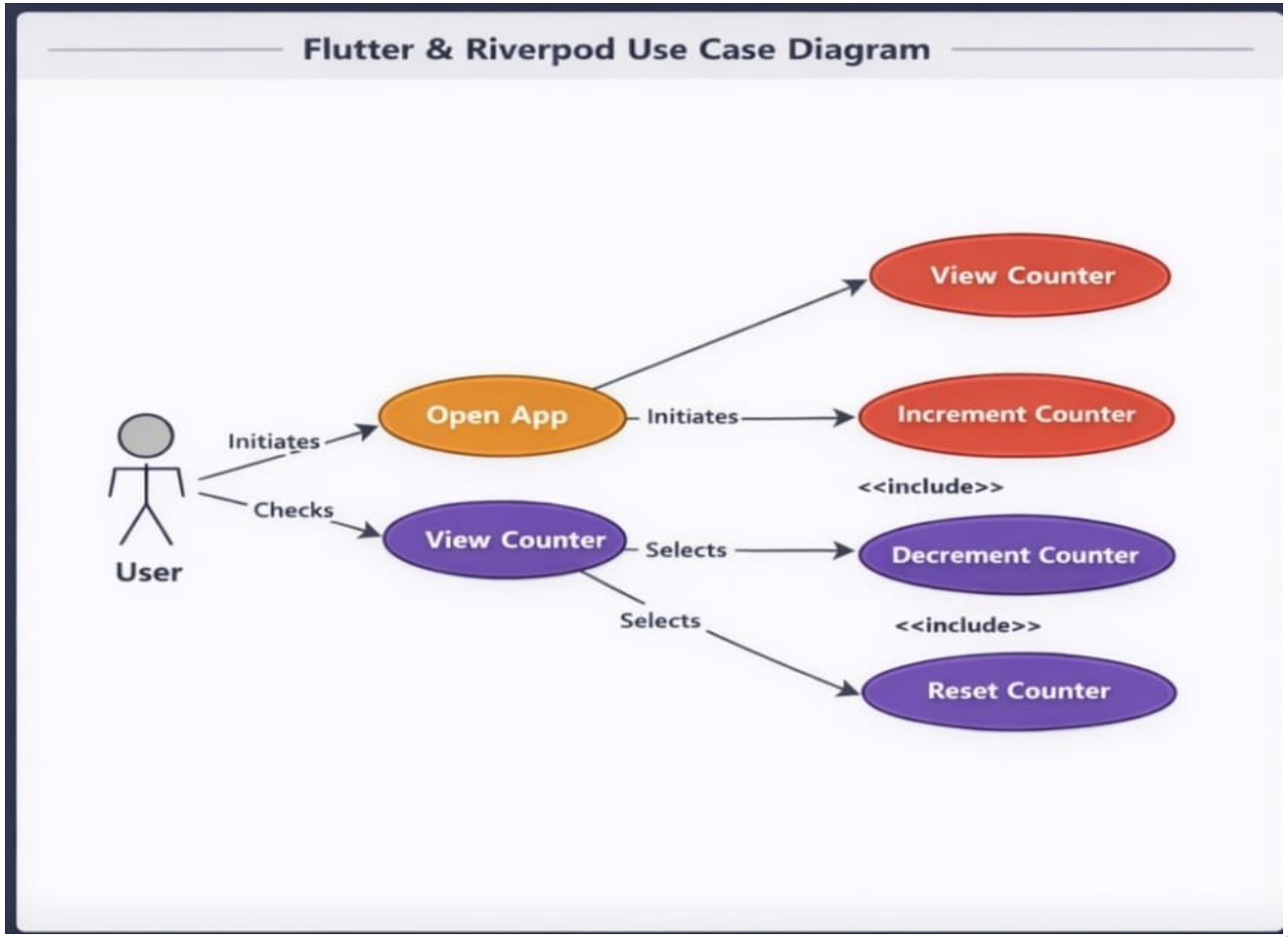
Data Structure:



UML Diagrams: Class Diagram



Usecase Diagram:



Features:

- The app has a ChatGpt-Style chat interface.
- Users can send messages and get replies instantly.
- The interface is simple and easy to use.
- It handles errors like network issues.
- It updates messages automatically on the screen.
- It is built using Flutter for mobile applications.
- Riverpod is used for State Management.

Advantages:

- It allows quick communication with AI.
- It provides a simple and user-friendly.
- It works on both Android and IOS platform.
- The system is scalable and easy to maintain.
- It improves user interaction experience.
- The UI design is clean and modern.

Conclusion:

- The project successfully creates a ChatGPT-style mobile application.
- The system is scalable and can be improved further.
- This project helps in understanding modern mobile app development.
- It can be extended with more advanced features in the future.